



BACHELOR'S THESIS TEMPLATE - TITLE OF PROJECT

Bachelor's Thesis

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Abstract: Replay-based continual learning is usually judged by the accuracy it reaches *after* each task. Continual evaluation has exposed a transient failure that this end-of-task view hides: the *stability gap*, a sharp drop in past-task accuracy at the start of every new task, before the model recovers. The gap persists even when the optimiser has access to the exact joint loss over all tasks, so it is not a defect of the objective but of the optimisation trajectory. We argue that, in its dominant components, the gap is a symptom of a single structural feature of online continual learning: at each task boundary the loss the optimiser follows changes discontinuously, from the replay loss alone to the sum of replay and current-task losses. We decompose the gap into three additive contributors — a first-step magnitude asymmetry, a finite-buffer estimator bias, and a trajectory effect — and identify the discontinuity as the source of the first and third. To remove it, we introduce the λ -*curriculum*, a homotopy that ramps the current-task loss weight from 0 to 1 over the first steps of each task, so the optimum moves along a continuous path rather than teleporting across the landscape. An envelope-theorem argument shows the replay loss is monotone non-decreasing along this path, bounding the trajectory contribution by $O(1/N)$ in the ramp length N . We add two refinements, a $\lambda_{\min}$ floor that restores early-task plasticity and momentum that suppresses the residual stochastic oscillation, and evaluate the method on rotated MNIST against vanilla and gradient-balanced Experience Replay.

1 Introduction

Deep neural networks learn rich representations from static, independent, and identically distributed data, but a vulnerability appears as soon as training becomes sequential: the network overwrites previously acquired knowledge to accommodate new data. This phenomenon, catastrophic forgetting [15, 17], is a central obstacle in machine learning. Standard approaches to accommodating new data frequently rely on offline retraining from scratch, a computationally expensive practice that remains impractical for systems requiring dynamic adaptation. Continual learning (CL) studies how to avoid this cost, and its core difficulty is the stability–plasticity dilemma: the model must stay plastic enough to learn new distributions while staying stable enough to preserve old ones [16].

Several families of techniques address forgetting, including replay, parameter and functional regularisation, optimisation-based methods, and template-based classification [19]. These methods reduce long-term forgetting, but they have historically been evaluated only at task boundaries or at coarse

intervals, which hides what happens within a task. Evaluating the network continuously exposes a failure these protocols miss. Lange et al. [13] did exactly this and identified the *Stability Gap*, a transient but severe collapse in past-task accuracy that occurs immediately upon the introduction of a new task. Even a well-tuned Experience Replay model that finishes a task with high accuracy on earlier tasks passes through a deep minimum during the first hundred steps of the new task.

This matters for three reasons. Deployed continual-learning systems are queried throughout training, so a model mid-transition commits errors its final checkpoint never would. The transience is itself a puzzle: the model clearly retains the knowledge needed to recover, so some mechanism temporarily disrupts and then restores it, and relearning what was already known wastes compute. Understanding that mechanism is a prerequisite for fixing the gap efficiently.

The discovery of the gap raised an immediate question: could it be removed by better approximating the joint loss over all tasks? Hess et al. [8] showed that the gap persists even under incremen-

tal joint training, where the model has access to all data from all tasks at every step. A perfect objective still produces transient forgetting, so the problem lies not in *what* is optimised but in *how* the optimisation proceeds.

To see why the trajectory itself produces the gap, consider the geometry of sequential optimisation. Figure 1.1 visualises the loss landscape for a transition from a learned task (Task A) to a new task (Task B). This is a two-dimensional simplification of a high-dimensional surface, but it captures the dynamics at play. The first three panels show Task A alone, Task B alone, and the joint loss $A + B$; the network starts at the converged minimum A of Task A, and the goal is to reach the joint minimum $A \cap B$. Running stochastic gradient descent (SGD) on the joint loss from A follows the green curve: pulled by the steepest gradients of the joint landscape, the trajectory bows outward, leaving Task A’s basin and crossing a region of higher Task-A error before converging at $A \cap B$. The dip on Task A along this arc is the stability gap, and work on linear mode connectivity [4, 5] shows it is avoidable: distinct SGD optima are often joined by simple paths of non-increasing loss, such as the dashed corridor from A to $A \cap B$.

The discontinuity at the task boundary is the source of this geometry. While Task A is learned the optimiser descends the replay loss alone; the instant Task B begins, the objective switches in a single step to the sum of the replay and current-task losses. The parameter does not move during the switch, but the surface the optimiser follows, and the location of its minimum, jumps from θ_0^* to θ_{joint}^* . We call this *landscape teleportation*: from the optimiser’s point of view the surface it was descending disappears and is replaced by another whose minimum lies elsewhere. The first step on the new surface is one-sided — the replay gradient is near zero at θ_0^* while the current-task gradient is not — so SGD is shoved out of Task A’s basin before the replay term can pull it back.

Given that the gap follows from a discontinuity, two paradigms address it. *Path-finding* methods accept the discontinuity and navigate the broken landscape carefully, constraining the SGD update with second-order information about the past task so that motion avoids directions in which the replay loss rises sharply [3, 10, 11, 14]. These methods treat the symptoms of the discontinuity; they do

nothing about the discontinuity itself. *Landscape-shaping* removes the discontinuity at its source. The λ -**curriculum** is a homotopy that ramps the current-task loss weight from 0 to 1 over the first N steps of each transition,

$$L_\lambda(t) = L_{\text{replay}} + \lambda(t) L_{\text{current}}, \quad \lambda(t) = \text{clip}\left(\frac{t}{N}, 0, 1\right), \quad (1.1)$$

so the optimum $\Theta^*(\lambda)$ traces a continuous curve from θ_0^* (at $\lambda = 0$) to θ_{joint}^* (at $\lambda = 1$). The model rides this moving valley rather than being teleported across an abyss (fourth panel of Figure 1.1). An envelope-theorem argument (Section 4) guarantees that the replay loss is monotone non-decreasing along $\Theta^*(\lambda)$, so in the continuous limit no overshoot is possible. The curriculum preserves the natural magnitudes of both gradients — it modulates only the *weight* on the current-task loss, leaving SGD’s step-size adaptation intact.

The basic linear ramp has two operational shortcomings, each met by a small refinement. First, a ramp of length N leaves λ near zero for the earliest steps, stalling progress on the new task; a λ_{min} **floor** keeps the current-task gradient present from the first step without re-introducing the depth that the curriculum removes. Second, the per-step accuracy curve retains a stochastic oscillation that the continuous-limit argument abstracts away; **momentum** acts as a low-pass filter that suppresses this mini-batch noise while preserving the slow drift along $\Theta^*(\lambda)$.

The remainder of this thesis develops these ideas. Section 2 reviews continual learning, Experience Replay, the path-finding methods we compare against, the stability-gap literature, and the related work on blurry task boundaries. Section 3 develops the discontinuity framing, decomposes the gap into three contributors, and contrasts path-finding with landscape-shaping. Section 4 states the research questions, and Section 5 describes the experimental testbed and the conditions used to test each prediction.

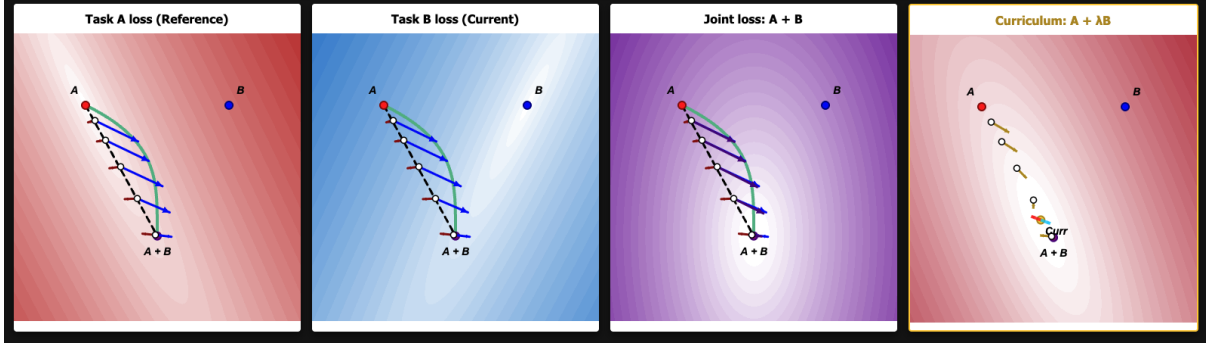


Figure 1.1: Conceptual visualisation of the optimisation landscape at a task transition, shown over a shared two-dimensional parameter space. Task A (red) and Task B (blue) are the single-task losses with minima at A and B ; the joint loss $A+B$ (purple) has its minimum at $A \cap B$. The dashed line is the straight interpolation from A to $A \cap B$; the green curve is the steepest-descent trajectory of the joint loss, which bows away from Task A’s basin and induces the stability gap. The fourth panel shows the λ -curriculum loss $A + \lambda B$: a small weight λ on Task B reshapes the basin into a valley that runs continuously from A toward the joint minimum, so the optimum moves along the corridor instead of teleporting across it. Arrows are normalised to a fixed length and show gradient direction only.

2 Background and Related Work

2.1 Continual Learning

Continual (or lifelong) learning trains a model on a non-stationary stream of tasks $T_0, T_1, T_2, \dots$ without joint access to all of them at once. Standard SGD on such a stream forgets catastrophically [15, 17]: gradient updates for a new task overwrite the weights that stored past knowledge, because the objectives are never minimised simultaneously. Three families of methods address this [19]. Regularisation-based methods add a penalty that slows updates to parameters important for past tasks, measured by a diagonal Hessian approximation [?]. Architecture-based methods allocate separate capacity to each task at the cost of a growing model. Replay-based methods keep a memory buffer of past examples and replay them during new-task training; this is the family studied here.

2.2 Experience Replay

Experience Replay (ER) is the simplest replay baseline [14]. At each step the model takes one forward–backward pass on the current-task mini-batch and one on a replay mini-batch sampled from the buffer,

then sums the two gradients before the optimiser step,

$$\theta \leftarrow \theta - \eta (\nabla_{\theta} L_{\text{current}}(\theta) + \nabla_{\theta} L_{\text{replay}}(\theta)). \quad (2.1)$$

This “standard” mode is used throughout the thesis; a “balanced” variant that normalises each gradient before summing is treated in Section 3 as a path-finding control. The buffer is populated by reservoir sampling [?], so every sample seen so far occupies a slot with equal probability regardless of arrival order. During T_0 the buffer is empty by construction, so ER reduces to plain SGD on the first task and the stability gap is measurable only from T_1 onward. ER’s simplicity makes it a strong, well-understood baseline, and because it does nothing to balance gradient magnitudes or directions it is the cleanest setting in which each contributor to the gap can be isolated.

A finite buffer is an imperfect estimator of the true past-task distribution. [?] frame buffer selection as a constraint-reduction problem in which the stored samples approximate the feasibility region defined by the full past data; reservoir sampling is one (non-optimal) policy within that frame. The replay gradient computed on a small buffer therefore differs in direction and magnitude from the gradient on all past data. This estimator gap is present at every step and disappears under no schedule; we

report it as a separate, persistent contributor to the stability gap.

2.3 Path-Finding Methods

A second class of replay-based and replay-adjacent methods navigates the discontinuous landscape at the task boundary by constraining the SGD update with second-order information about the past tasks. GEM projects the new-task gradient onto the nearest direction that does not increase the loss on any past task in the buffer [14], and A-GEM approximates this with a single averaged past gradient to make it cheap [3]. Both constrain the update direction at the current step but enforce the constraint step-wise, not across the trajectory. EWC adds a Fisher-weighted L_2 penalty around the past optimum, slowing updates in directions the old task constrained [?]. NCL preconditions the update by the precision matrix of the past-task loss, producing a natural-gradient step that respects the curvature of the replay loss [10]. In the framing of this thesis (Section 3), all of these are *path-finding* methods: they accept the discontinuity at the task boundary and invest second-order machinery in picking a safe trajectory across the teleported terrain.

2.4 The Stability Gap

Lange et al. [13] characterise the stability gap as the transient accuracy dip on previously seen tasks at the start of new-task training in replay-based methods. They observe that the gap is largest in the first few hundred steps of each task, that larger buffers produce smaller gaps, that recovery is partial with the minimum accuracy systematically below the pre-task baseline, and that the gap survives reservoir sampling and moderate learning rates. Hess et al. [8] add a mechanistic reading: at a task optimum the past-task gradient is near zero, so the first step of new-task training is one-sided, and the geometry of the joint-loss landscape at the transition predicts gap severity. Subsequent work has shown the gap to be broad rather than incidental: it appears in joint incremental learning of homogeneous tasks [9], in continual pre-training of large models [6], and in relation to the classification head [20], and various mitigations have been proposed, from temporal ensembles [18] to explicit gap-reduction procedures [7].

2.5 The Gap Survives the Joint-Loss Oracle

A theoretical anchor for this thesis comes from Kao et al. [10], who isolate a clean fact about gradient descent through a task transition. Remove every replay confound — no buffer, no sampling noise, no magnitude imbalance — and give the optimiser exact access to the joint loss $L_{\text{joint}} = L_{T_0} + L_{T_1}$, then run SGD from a stationary point θ_0^* of L_{T_0} . This is the strongest possible replay regime, equivalent to multi-task training with perfect knowledge of the past. Even so, accuracy on T_0 dips before recovering. At θ_0^* the past-task gradient is near zero while the new-task gradient is not, so the first step moves essentially along $-\nabla L_{T_1}(\theta_0^*)$; a second-order expansion gives

$$L_{T_0}(\theta_1) - L_{T_0}(\theta_0^*) \approx \frac{1}{2} \eta^2 \nabla L_{T_1}(\theta_0^*)^\top H_{T_0}(\theta_0^*) \nabla L_{T_1}(\theta_0^*) > 0, \quad (2.2)$$

whenever H_{T_0} has positive curvature in the direction of ∇L_{T_1} , which it generically does at a strict minimum. The dip is not caused by bad gradients — the gradient is exact — but by the *trajectory* the optimiser takes. Section 3 reproduces this as the cleanest isolation of the trajectory contributor and shows that a continuous schedule removes it in the continuous-time limit.

2.6 Blurry Task Boundaries

A parallel line of work relaxes the assumption of abrupt task boundaries [1, 2, 12]. In blurry or boundary-free settings the stream does not switch from T_0 to T_1 at a single step; instead T_1 samples are introduced gradually, so at time t the minibatch is drawn from a mixture $(1 - \alpha(t)) P_{T_0} + \alpha(t) P_{T_1}$ with $\alpha(t)$ rising from 0 to 1. The expected SGD update is then $\alpha \nabla L_{T_1}(\theta) + (1 - \alpha) \nabla L_{T_0}(\theta)$, a weighted sum of the two task gradients. Substituting $\lambda = \alpha/(1 - \alpha)$ shows this is, up to an overall scale, the gradient of $L_{T_0} + \lambda L_{T_1}$ — the same loss the λ -curriculum descends. Both interventions implement the same fix to the one-sided first step: they keep the past-task gradient present at full strength from step one while introducing the new-task gradient gradually. The two differ in scope. Blurry boundaries achieve the mixing through the data stream and need no buffer, but lose the past-task signal entirely once $\alpha = 1$; the λ -curriculum is

built on replay and retains an explicit replay signal at $\lambda = 1$. The curriculum is also the cleaner controlled experiment, since it modifies one weight in the loss while data sampling and replay magnitude are held fixed, isolating which ingredient of a gradual transition reduces the gap.

3 Motivation: The Gap as a Symptom of a Discontinuous Loss

This section develops the argument that the stability gap is, in its dominant components, the downstream symptom of one structural feature of on-line continual learning: the loss the optimiser follows changes discontinuously at each task boundary. Section 3.1 states the homotopy that the λ -curriculum implements. Section 3.2 decomposes the gap into three additive contributors and identifies which the curriculum addresses. Section 3.3 contrasts path-finding with landscape-shaping, and Section 3.4 states the envelope-theorem bound on the trajectory contributor.

3.1 The Homotopy Family L_λ

Standard online continual learning trains on a sequence of objectives that change discontinuously. During T_0 the optimiser descends $L_{\text{replay}} = L_{T_0}$ alone. The moment T_1 begins, the objective switches in a single step to $L_{\text{replay}} + L_{\text{current}} = L_{T_0} + L_{T_1}$. The parameter θ does not move during the switch, but the surface the optimiser follows — and the location of its minimum — jumps from θ_0^* to θ_{joint}^* . The λ -curriculum replaces this jump with a one-parameter family of losses,

$$L_\lambda(\theta) = L_{\text{replay}}(\theta) + \lambda L_{\text{current}}(\theta), \quad \lambda \in [0, 1], \quad (3.1)$$

where $\lambda = 0$ recovers pure replay (optimum at $\theta_0^* := \arg \min L_{\text{replay}}$) and $\lambda = 1$ recovers the joint loss (optimum at θ_{joint}^*). Ramping λ along a continuous schedule defines two trajectories: the *optimum path* $\Theta^*(\lambda) := \arg \min_\theta L_\lambda(\theta)$, a continuously differentiable curve from θ_0^* to θ_{joint}^* under standard regularity assumptions, and the *SGD trajectory* $\theta(t)$, the actual sequence of iterates. The fourth panel of Figure 1.1 shows the reshaped basin for a small λ .

3.2 Three-Contributor Decomposition

Consider the first vanilla-ER step at the start of T_1 . Because θ_0^* is a stationary point of L_{replay} , the true replay gradient there is near zero, and the update is

$$\theta_1 \approx \theta_0^* - \eta (\varepsilon(\theta_0^*) + g_{\text{new}}(\theta_0^*)), \quad (3.2)$$

where $g_{\text{new}} := \nabla L_{\text{current}}$ and ε is the zero-mean finite-buffer estimator error. This step has three features, each driving a distinct contribution to the subsequent dip on T_0 .

Magnitude asymmetry (G_{mag}). The new-task gradient $g_{\text{new}}(\theta_0^*)$ is generically $O(1)$, while ε is mean-zero and $O(1/\sqrt{k})$ in the buffer size k . The first step is therefore dominated by $-\eta g_{\text{new}}(\theta_0^*)$, and a second-order expansion of the replay loss gives

$$L_{\text{replay}}(\theta_1) - L_{\text{replay}}(\theta_0^*) \approx \frac{1}{2} \eta^2 g_{\text{new}}^\top H_{\text{replay}}(\theta_0^*) g_{\text{new}} > 0, \quad (3.3)$$

whenever H_{replay} has positive curvature along g_{new} . This is the dominant contributor and a direct consequence of the discontinuity.

Estimator bias (G_{est}). On any individual run $\varepsilon \neq 0$ and adds a stochastic kick. Following [?], a finite buffer is a finite-sample estimator of the past data, and the per-step error contributes a depth that does not vanish under any schedule because it is independent of the loss composition. This contributor is reported but is not the one the curriculum is designed to fix.

Trajectory effect (G_{traj}). Even with the exact joint-loss gradient and no estimator error, the geometric observation of Kao et al. [10] (Section 2.5) remains: an instantaneous switch from $\lambda = 0$ to $\lambda = 1$ lands the iterate off $\Theta^*(1)$, where the replay loss is strictly higher than at the joint optimum. This is the contributor the envelope-theorem argument addresses.

The contributors act at different characteristic steps — G_{mag} at step one, G_{est} across all steps, G_{traj} over the first equilibration window — which lets an ablation isolate each empirically (Section 5).

3.3 Path-Finding versus Landscape-Shaping

Methods can be organised by what they do with the discontinuity rather than by what curvature information they use. *Path-finding* accepts the discontinuity: the objective still jumps at the boundary, and the intervention reshapes *how* the optimiser moves, replacing plain SGD with a preconditioned step $\theta_{t+1} = \theta_t - \eta M(\theta_t) \hat{g}_t$. Here M is a second-order quantity — the inverse Fisher of EWC [?], the precision matrix of NCL [10], or the feasible-set projection of GEM [14]. Gradient balancing, the “balanced” ER mode, is path-finding too: it leaves the schedule discontinuous but rescales the gradient so the first step is not dominated by g_{new} .

Landscape-shaping removes the discontinuity. It modifies the objective in a neighbourhood of the boundary so the surface the optimiser descends is continuous through the transition, leaving the geometry of the step to SGD’s natural step-size adaptation. The λ -curriculum is the canonical example: $\hat{g}_t = \nabla(L_{\text{replay}} + \lambda(t) L_{\text{current}})(\theta_t)$ with $\lambda(t)$ a continuous schedule from 0 to 1. Three differences follow. Preconditioners depend on the surface they are built over — NCL’s factors are computed on linear weights only — whereas $\lambda(t)$ multiplies the loss and flows through the backward pass to every parameter. Any preconditioner that normalises the gradient breaks SGD’s step-size adaptation, the source of the accuracy cost incurred by gradient balancing, while landscape-shaping leaves it intact. And the preconditioned step requires a per-step matrix–vector product, whereas the curriculum adds one scalar multiply per backward pass.

3.4 The Envelope-Theorem Bound

The reason a continuous schedule removes the trajectory contributor is that the replay loss cannot rise along the optimum path. At the optimum the first-order condition is $\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0$. Differentiating it implicitly in λ and substituting into the chain rule for $\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda))$ yields

$$\frac{d}{d\lambda} L_{\text{replay}}(\Theta^*(\lambda)) = \lambda \nabla L_{\text{current}}^\top H_\lambda^{-1} \nabla L_{\text{current}} \geq 0, \quad (3.4)$$

since $\lambda \geq 0$ and $H_\lambda = H_{\text{replay}} + \lambda H_{\text{current}} \succ 0$ along the path. The replay loss is monotone non-

decreasing along $\Theta^*(\lambda)$, so no overshoot is possible on the optimum path. An adiabatic-tracking argument shows the SGD iterates stay within $O(1/N)$ of the optimum path for ramp length N , so

$$\sup_t L_{\text{replay}}(\theta(t)) \leq L_{\text{replay}}(\theta_{\text{joint}}^*) + O(1/N). \quad (3.5)$$

The first term is the unavoidable cost of joint training; the $O(1/N)$ term is the trajectory contributor that the curriculum eliminates as N grows. This is the prediction tested in the experiments: under the linear curriculum, gap depth should fall as N grows, while joint-task accuracy stays close to vanilla ER because the step-size adaptation is preserved. Two refinements act on errors the continuous-limit argument abstracts away. A $\lambda_{\min}$ floor, $\lambda(t) = \max(\lambda_{\min}, t/N)$, starts the homotopy at an interior point where the current-task gradient is already present, trading a slightly higher gap floor for faster early progress on T_1 . Momentum averages the per-step gradient over recent steps, suppressing the stochastic oscillation that survives the curriculum without changing where the gradient points; it therefore cannot substitute for the curriculum, since the one-sided first step is deterministic.

4 Research Questions

The framing of Section 3 makes a set of falsifiable predictions about how the λ -curriculum and its refinements should behave. We evaluate them on rotated MNIST with a small multilayer perceptron, the regime in which the envelope-theorem bound is sharply testable, using vanilla and gradient-balanced Experience Replay as baselines. Five research questions structure the experiments.

RQ1 (linear curriculum). Does the linear λ -curriculum applied to standard ER monotonically reduce stability-gap depth as the ramp length N grows, and does it do so without the final-accuracy cost incurred by gradient balancing?

RQ2 (adaptive curriculum). Does an adaptive λ -schedule, which sets λ from the live ratio of replay-to-current gradient magnitudes and so removes the explicit knob N , reduce gap depth without that final-accuracy cost?

RQ3 (momentum). Does adding momentum on top of the curriculum reduce the residual depth fluctuation predicted by the discrete-time stochastic argument? And does momentum on its own, without the curriculum, produce any comparable benefit — the prediction being that it does not, since the one-sided first step is deterministic?

RQ4 ($\lambda_{\min}$). Does a small floor $\lambda_{\min} > 0$ on the adaptive schedule recover early-task learning speed without re-introducing the depth pathology the curriculum was designed to remove?

RQ5 (decomposition). How does the gap depth decompose into the magnitude, estimator, and trajectory contributions of Section 3.2, and what does each share reveal about why a curriculum is the right shape of fix?

Section 5 describes the testbed, the metrics used to measure gap depth and final accuracy, and the full set of conditions that isolate each contributor and test each prediction.

5 Methods

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